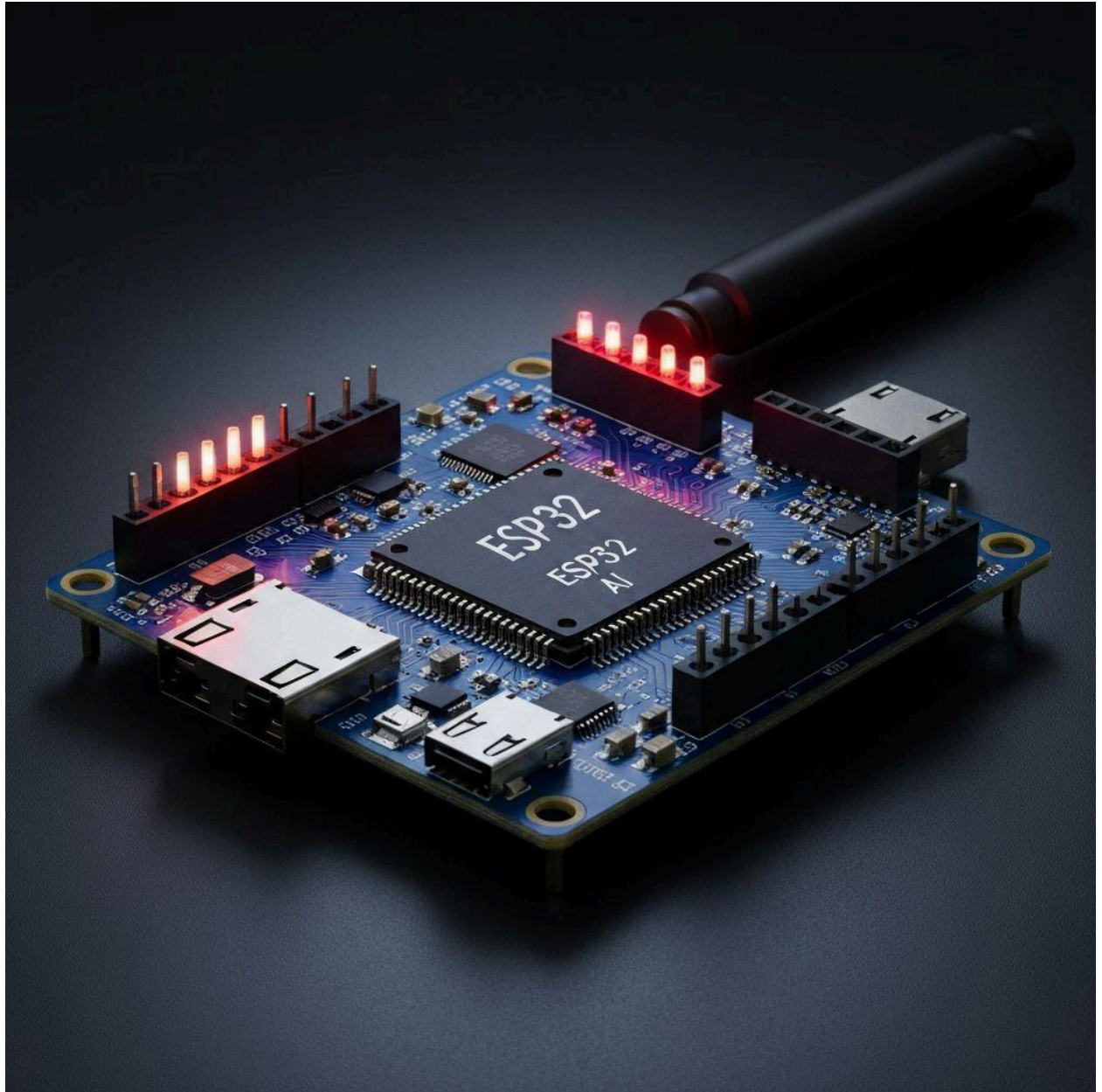
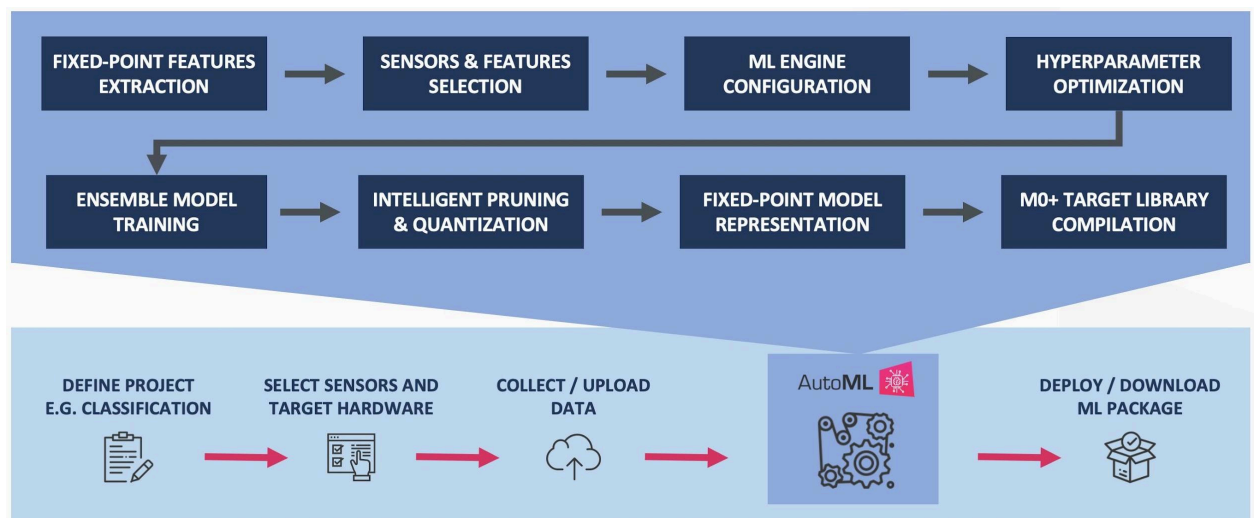
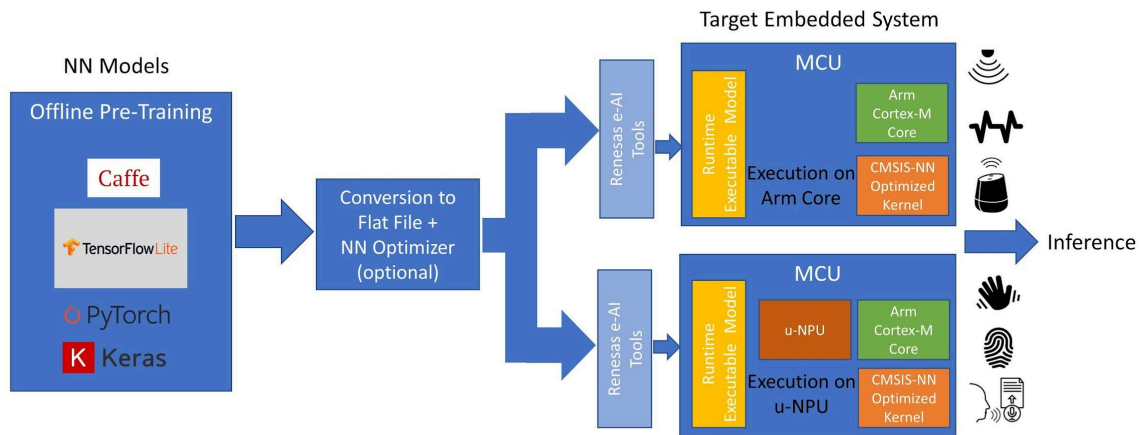


Embedded AI Engineer Roadmap (2026)









An **Embedded AI Engineer** builds AI systems that run directly on devices such as microcontrollers, edge computers, robots, cameras, drones, and IoT systems instead of relying entirely on cloud servers.

Skills to Learn

1. Programming

- Python
- C
- C++
- Embedded Linux

2. Electronics

- GPIO
- I2C
- SPI
- UART
- PWM
- ADC/DAC

3. Hardware Platforms

Raspberry Pi Raspberry Pi 5

Espressif Systems ESP32 / ESP32-CAM

NVIDIA Jetson Nano / Orin

4. AI & Computer Vision

- OpenCV
- Deep Learning
- CNN
- Object Detection
- Image Classification
- Tracking

5. Edge AI

- TensorFlow Lite
 - TensorFlow Lite Micro
 - ONNX Runtime
 - TinyML
-

Software Stack

AI Libraries

```
pip install numpy
```

```
pip install pandas
```

```
pip install opencv-python
```

```
pip install tensorflow
```

```
pip install torch
```

```
pip install ultralytics
```

Embedded Development

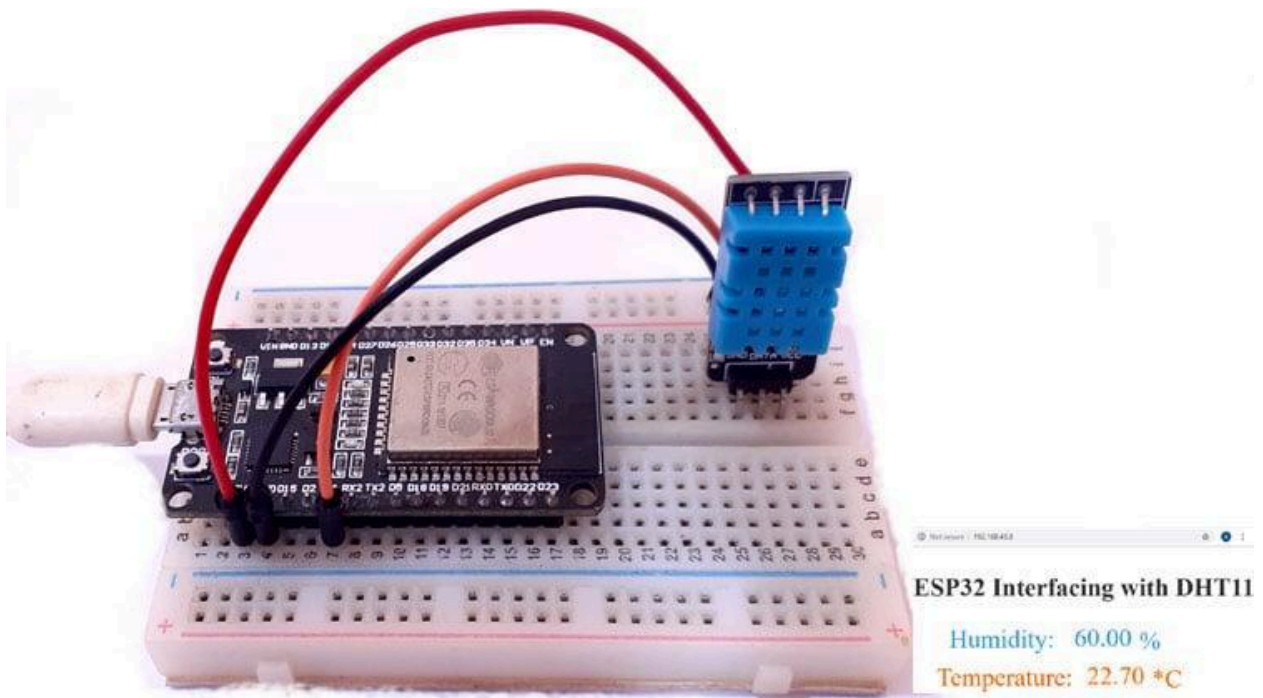
```
sudo apt install git cmake build-essential
```

ESP32

Install:

- Arduino IDE
 - ESP-IDF
-

Beginner Projects



ESP32 Environmental Monitor

Not Connected

TEMPERATURE

-- °C

Waiting...

HUMIDITY

-- %

Waiting...

PM2.5 LEVEL

-- µg/m³

Waiting...

CO2 CONCENTRATION

-- ppm

Waiting...

NOISE LEVEL

-- dB

Waiting...

DUST CONCENTRATION

-- µg/m³

Waiting...

RELAY STATUS

--

Standby

Sensor History Graph

Control Panel

Operating Mode

Select Mode

Automatic Mode

Auto: System controls relay based on thresholds
Manual: You control the relay directly

Manual Relay Control

Turn ON

Turn OFF

Manual control is only available in Manual mode

Auto Mode Thresholds

Temperature (°C)

30

PM2.5 (µg/m³)

50

CO2 (ppm)

800

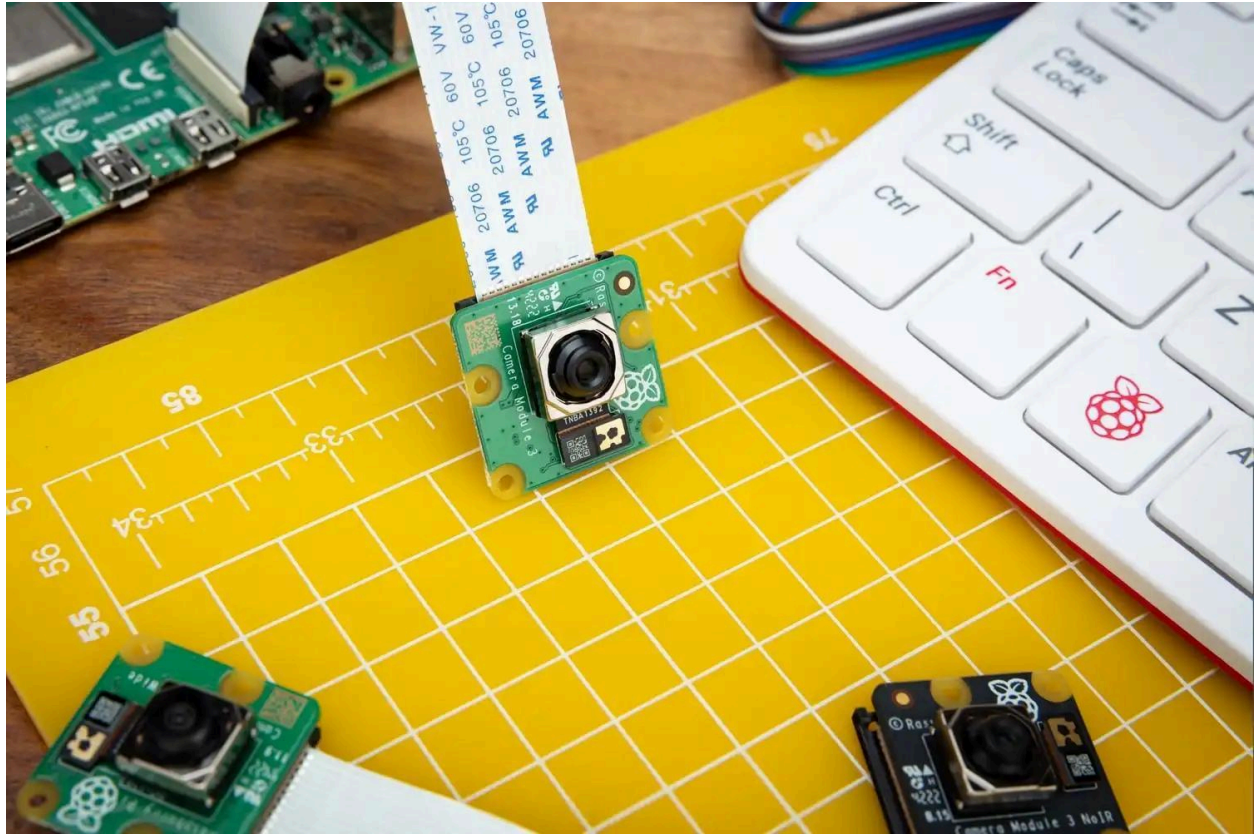
Noise (dB)

70

Save Thresholds

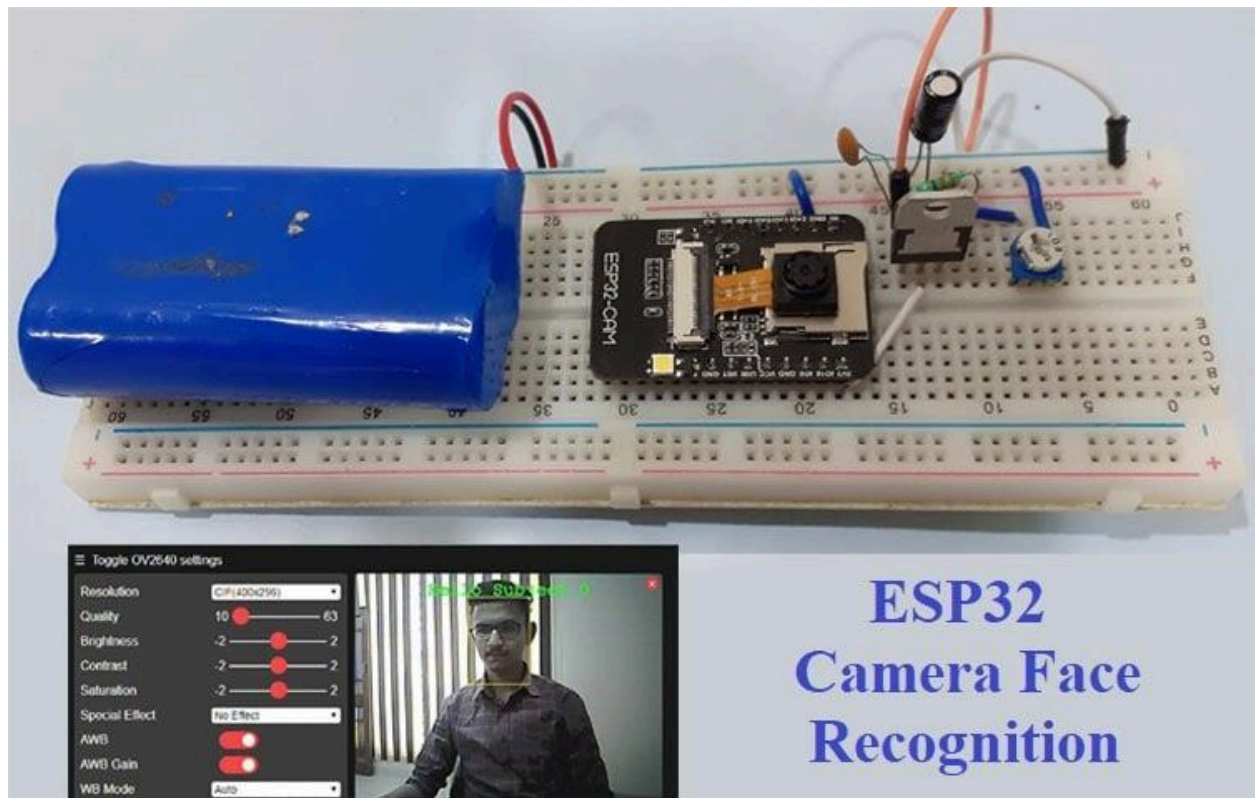
Device ID: esp32_env01

Last update: Waiting for data...



1. Temperature Monitoring
 2. Smart Home Automation
 3. Motion Detection Camera
 4. IoT Sensor Dashboard
 5. Face Detection Camera
-

Intermediate Projects

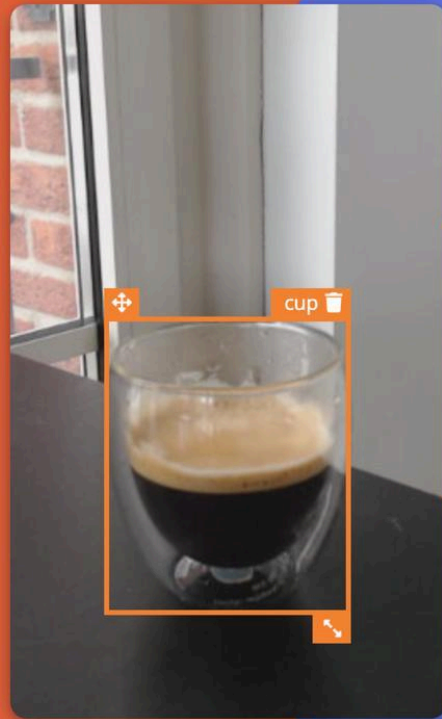




	Human Vision	Computer Vision
Input		
Sensing Device	*eye photo	*cctv camera photo
Proceesing Device	*brain photo	*computer with ml model photo
Output	Person , car , Tree	Person , car , Tree

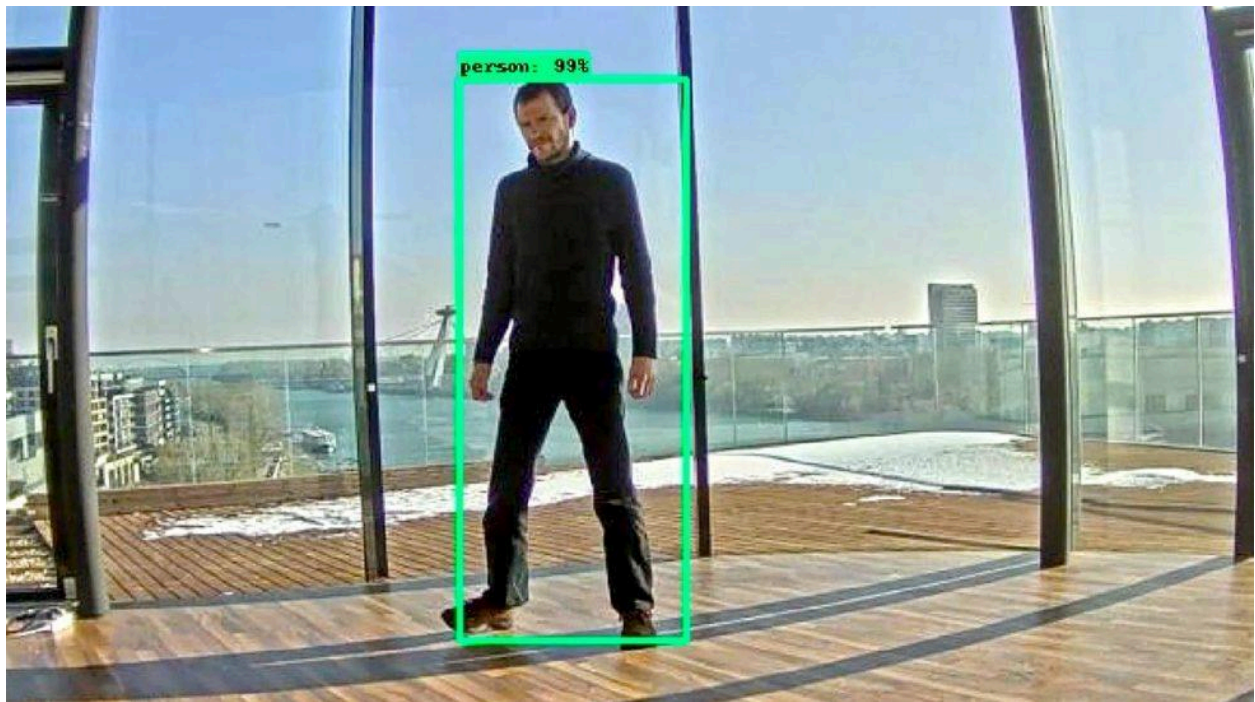
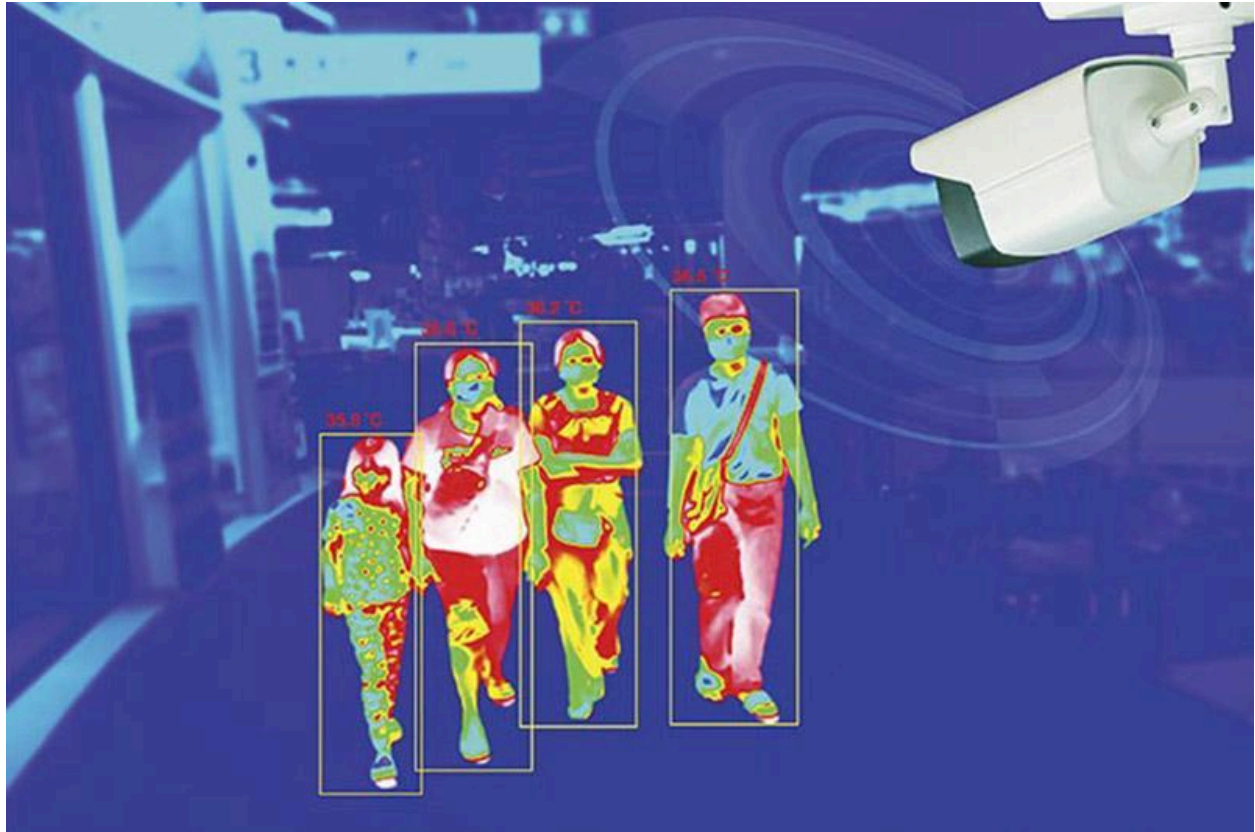
NEW FEATURE

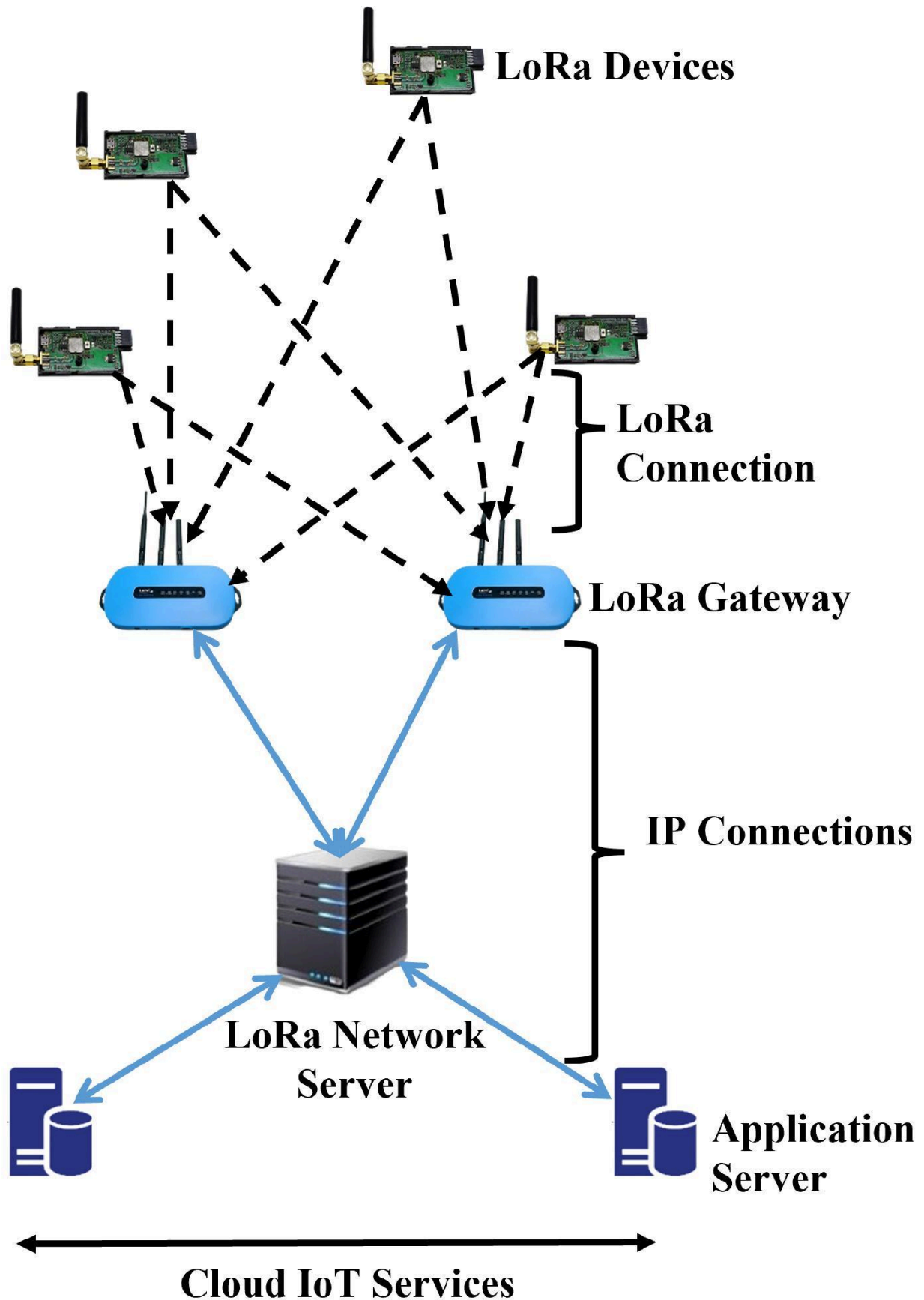
AI-Assisted Labeling for Object Detection



1. Face Recognition Door Lock
2. AI Attendance System
3. Smart Parking System
4. Vehicle Counter
5. Fire Detection System

Advanced Projects







Raspberry Pi 5 + MLX90640 Thermal AI

Features:

- Human Detection
- Temperature Mapping
- Intrusion Alerts

Raspberry Pi 5 + YOLO CCTV

Features:

- Person Detection
- Vehicle Detection
- Real-Time Tracking

ESP32-CAM AI Surveillance

Features:

- Face Detection
- Object Detection
- Wi-Fi Streaming

LoRa AI Sensor Network

Using:

- Raspberry Pi 5
- SX1262 LoRa
- ESP32 Nodes

Applications:

- Agriculture
 - Wildlife Monitoring
 - Smart City Sensors
-

Important Technologies

TinyML

Run AI models directly on:

- ESP32
- STM32
- Arduino

Learn:

- TensorFlow Lite Micro
- Quantization
- Model Optimization

Edge AI

Run AI locally on:

- Raspberry Pi
- Jetson
- Industrial Devices

Benefits:

- Low Latency
 - Offline Operation
 - Better Privacy
-